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Research Article

Thermal Transport across Metal/β-Ga_2O_3 Interfaces

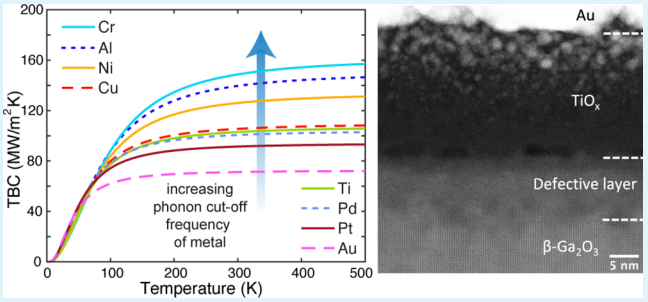
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ABSTRACT: In this work, we study the thermal transport at β-Ga_2O_3/metal interfaces, which play important roles in heat dissipation and as electrical contacts in β-Ga_2O_3 devices. A theoretical Landauer approach was used to model and elucidate the factors that impact the thermal transport at these interfaces. Experimental measurements using time-domain thermoreflectance (TDTR) provided data for the thermal boundary conductance (TBC) between β-Ga_2O_3 and a range of metals used to create both Schottky and ohmic electrical contacts. From the modeling and experiments, the relation between the metal cutoff frequency and the corresponding TBC is observed. Moreover, the effect of the metal cutoff frequency on TBC is seen as the most significant factor followed by chemical reactions and defects between the metal and the β-Ga_2O_3. Among all β-Ga_2O_3/metal interfaces, for Schottky contacts, Ni/β-Ga_2O_3 interfaces show the highest TBC, while for ohmic contacts, Cr/β-Ga_2O_3 interfaces show the highest TBC. While there is a clear correlation between TBC and the phonon cutoff frequency of metal contacts, it is also important to control the chemical reactions and other defects at interfaces to maximize the TBC in this system.



KEYWORDS: *gallium oxide, thermal management, interface, thermal boundary conductance, time-domain thermoreflectance, metal/β-Ga_2O_3*

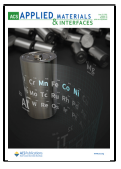
1. INTRODUCTION

β-Ga_2O_3 is an ultrawide bandgap (UWBG) semiconductor with properties that are attractive to create advanced power electronics and radio frequency devices.[1−3] However, one issue that consistently arises with β-Ga_2O_3 is the fact that its thermal conductivity (ranging from 11 to 27 W/(m K) at room temperature depending on the crystallographic orientation[4−7]) is low and can lead to high device thermal resistances. The low thermal conductivity of β-Ga_2O_3 will limit the power density in β-Ga_2O_3 power electronic devices due to concerns over the operating temperature and reliability of devices. Therefore, it is significant to understand the thermal transport inside β-Ga_2O_3 and at interfaces to develop advanced cooling strategies for β-Ga_2O_3 devices.

The development of β-Ga_2O_3 devices will take on both lateral and vertical architectures. In lateral architectures, current flows laterally from the source to drain and heating is biased to the top side of the device architecture as shown in Figure 1a. In a recent study of cooling strategies for lateral β-Ga_2O_3 devices,[8] it is estimated that, with double-side cooling strategies, these devices can work within reasonable temperature limits and achieve high power densities. In this approach, devices are created from β-Ga_2O_3 integrated with high-thermal-conductivity substrates and with metal contacts being used as pathways for heat dissipation on both sides. To optimize the thermal dissipation efficiency in β-Ga_2O_3 devices, the thickness of the β-Ga_2O_3 layer would be thin because of its low thermal conductivity. As a result, the thermal boundary conductances (TBCs) across β-Ga_2O_3/metal and β-Ga_2O_3/substrate interfaces play important roles in the total thermal resistances of devices and should be greater than 100 MW/(m^2 K). For vertical devices as shown in Figure 1b, heat flow occurs through metal contacts into heat spreaders or power electronic substrates also play a critical role in the heat dissipation path. Therefore, it is important to understand and predict the thermal boundary conductance at β-Ga_2O_3/metal interfaces as a guide for the thermal design of β-Ga_2O_3 devices.

Due to the importance of β-Ga_2O_3/metal interfaces, there have been several studies concerning the thermal boundary conductances of Au/β-Ga_2O_3, Ti/β-Ga_2O_3, Al/β-Ga_2O_3, and Cr/β-Ga_2O_3 interfaces.[7,9,10] Previous studies have shown that the TBC value varies at different β-Ga_2O_3 interfaces. However, the choice of metal contacts must also consider the dual roles of thermal and electrical contacts (Schottky and ohmic). Thus,

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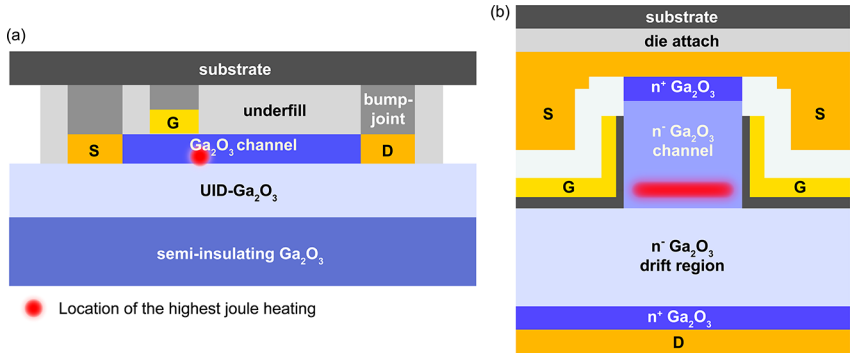


Figure 1. Schematic of (a) a lateral β-Ga_2O_3 metal-semiconductor field-effect transistor (MESFET) and (b) a vertical β-Ga_2O_3 metal-oxide field-effect transistor (MOSFET). S, D, and G represent the source, drain, and gate, respectively. Locations of the highest joule heating are highlighted in red.

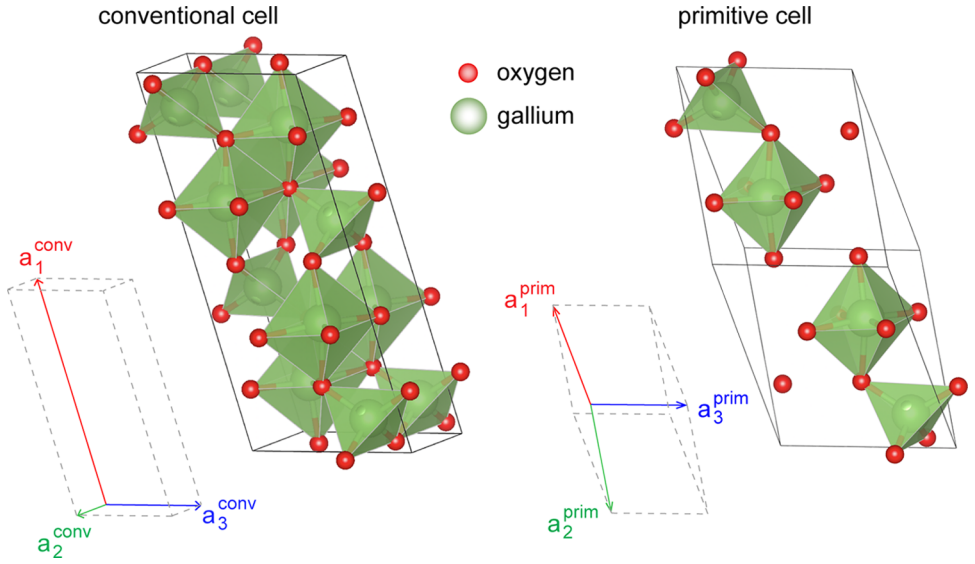


Figure 2. Structures of the β-Ga_2O_3 conventional unit cell and primitive cell. The lattice vectors of both cells are shown as $\mathbf{a}_1^{conv}$, $\mathbf{a}_2^{conv}$, $\mathbf{a}_3^{conv}$ and $\mathbf{a}_1^{prim}$, $\mathbf{a}_2^{prim}$, $\mathbf{a}_3^{prim}$. The relations between these lattice vectors are $\mathbf{a}_1^{prim} = \frac{(\mathbf{a}_1^{conv} + \mathbf{a}_2^{conv})}{2}$, $\mathbf{a}_2^{prim} = \frac{(-\mathbf{a}_1^{conv} + \mathbf{a}_2^{conv})}{2}$, and $\mathbf{a}_3^{prim} = \mathbf{a}_3^{conv}$.

a study involving a wider range of metals that fulfill these dual roles is important to predict how they impact the TBC with β-Ga_2O_3. Moreover, the use of models to help describe the TBC variation with β-Ga_2O_3 is lacking but needed to help guide our understanding of metal/β-Ga_2O_3 contacts and select the optimal metals as ohmic and Schottky contacts in devices to minimize the total resistance. In this work, we apply the Landauer approach to study the TBC at β-Ga_2O_3/metal interfaces, where the metal can be used as ohmic or Schottky contacts. For β-Ga_2O_3/metal interfaces, Au, Pt, Ni, Pd, and Cu are selected as representative Schottky contact metals,[11] and Ti, Cr, and Al are selected as ohmic contact metals.[12] While Al has not been applied as an ohmic contact in β-Ga_2O_3 devices, it has been used as ohmic contact for n-type GaN devices and is included in our study of β-Ga_2O_3/metal interfaces due to the fact that there are preexisting data for it.[11,13,14] Time-domain thermoreflectance (TDTR)[15−20] is applied to measure the TBC to compare with the modeling results, and scanning transmission electron microscopy (STEM) is applied to understand the interfacial structures. This work provides insight into the understanding of thermal transport at β-Ga_2O_3 interfaces and sheds light on the thermal design of β-Ga_2O_3 devices.

2. METHODOLOGY

To understand the heat transfer across β-Ga_2O_3/metal interfaces, a theoretical model is needed to capture the energy carrier transport process. The Landauer approach,[21−23] which was originally derived for the calculation of electrical current, is applied here to calculate the thermal boundary conductance G at β-Ga_2O_3/metal interfaces with phonons as the energy carriers. The idea of the Landauer formula is that the total heat flux q_1 from one side of the interface to the other can be calculated by integrating the product of energy density and carrier velocity over all frequencies. With the heat flux q_1 and q_2 from each side of the interface, the TBC value can be obtained from the net flux $q_{1^-2} = q_1 - q_2$ over the temperature drop ΔT as $q_{1^-2}/\Delta T$. At β-Ga_2O_3 interfaces, the general form of the Landauer approach is

$$G = \sum_{p_1} \frac{1}{2} \iint \hbar\omega D_1(\omega) v_1(\omega) \tau_{12}(\theta, \omega) \frac{df_{BE}}{dT} \cos\theta \sin\theta \, d\theta d\omega \quad (1)$$

where the sum Σ_p is over all phonon branches, D is the phonon density of states (DOS), f_{BE} is the Bose−Einstein distribution

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function to describe phonons, $\hbar$ is the reduced Planck constant, ω is the phonon angular frequency, v_1 is the phonon group velocity of material 1 (Ga_2O_3), τ_{12} is the transmission coefficient from material 1 to material 2 (in our case, from β-Ga_2O_3 to the metal), θ is the angle of incidence, and the sum is over all incident phonon branches. The diffuse mismatch model (DMM) is applied to calculate the spectral phonon transmission function,[23] and the calculation process can be found in the Supporting Information. Since the transmission function from DMM is spectral but not modal and not dependent on the incident angle, the Landauer formula of eq 1 can be simplified as

$$G = \sum_{p_1} \frac{1}{4} \int \hbar\omega D_1(\omega) v_1(\omega) \tau_{12}(\omega) \frac{df_{BE}}{dT} d\omega \quad (2)$$

The second-order interatomic force constants of β-Ga_2O_3 are obtained from density functional theory (DFT) with the Vienna ab initio simulation package (VASP),[24,25] and the phonon dispersion curves along a certain direction are calculated with Phonopy.[26] For metals, the phonon dispersion curves are obtained from experimental measurements.[27−29] With the DFT and experimental dispersion relation, the phonon group velocities v can be calculated as the slope of phonon dispersion curves, and the DOS can be calculated from the phonon wavevector and group velocities.

In our work, the β-Ga_2O_3 $(\bar{2}01)$ orientation surface is selected as the interface between β-Ga_2O_3 and metals because it is the most commonly used surface in β-Ga_2O_3 devices from literature studies.[11,12] For metals, the [100] orientation is selected as the direction perpendicular to the interface for face-centered cubic (FCC) metals in our study (Au, Pt, Ni, Pd, Cu, Cr, Al), and the [110] direction is used for hexagonal closed-packed (HCP) Ti. To correctly capture the phonon properties along specific orientations between certain high-symmetry points, it is important to thoroughly understand the β-Ga_2O_3 lattice structure, considering that β-Ga_2O_3 has a complex monoclinic crystal with the space group $C2/m$. It should be noted that the conventional unit cell of β-Ga_2O_3 (20 atoms) is different from its primitive cell (10 atoms)[30] as shown in Figure 2, but the plane indices and direction vectors of the conventional cell and primitive cell are sometimes mixed up. It is because the modeling of β-Ga_2O_3 like DFT calculations usually uses the smaller primitive cell to lower the computational cost, while fabrication and characterization usually refer to the conventional cell as tradition.

Here, the $(\bar{2}01)$ orientation refers to the $(\bar{2}01)$ plane of the β-Ga_2O_3 conventional cell, and it is orthogonal to $-2\boldsymbol{b}_1^{conv} + \boldsymbol{b}_3^{conv}$ from definition, where $\boldsymbol{b}_1^{conv}$, $\boldsymbol{b}_2^{conv}$, and $\boldsymbol{b}_3^{conv}$ are the basis of the reciprocal lattice vectors of the conventional cell. Also, for the conventional monoclinic β-Ga_2O_3 cell, $\mathbf{a}_1^{conv} \neq \mathbf{a}_2^{conv} \neq \mathbf{a}_3^{conv}$, and $\beta^{conv} \neq 90°$, and as a result, the $[\bar{2}01]$ vector is not orthogonal to the $(\bar{2}01)$ plane unlike cubic silicon and other crystals with mutually orthogonal direct lattice vectors.

In our work, the primitive cell is applied in the DFT calculations of the β-Ga_2O_3 harmonic properties, so the conventional indices are converted to the indices of the primitive cell based on the relation between the conventional and primitive direct lattice vectors and some coordinate transformation. The $(\bar{2}01)$ plane is in fact the $(\bar{1}11)$ plane of the primitive cell because $-2\boldsymbol{b}_1^{conv} + \boldsymbol{b}_3^{conv}$ and $-\boldsymbol{b}_1^{prim} + \boldsymbol{b}_2^{prim} + \boldsymbol{b}_3^{prim}$ have the same direction. Hence, we mainly consider the phonon dispersion from Γ to L in the first Brillouin zone corresponding to the $-\boldsymbol{b}_1^{prim} + \boldsymbol{b}_2^{prim} + \boldsymbol{b}_3^{prim}$ direction,[30,31] and the phonon dispersion curves of β-Ga_2O_3 from Γ to L are shown in Figure 3. In our modeling, the phonon properties of

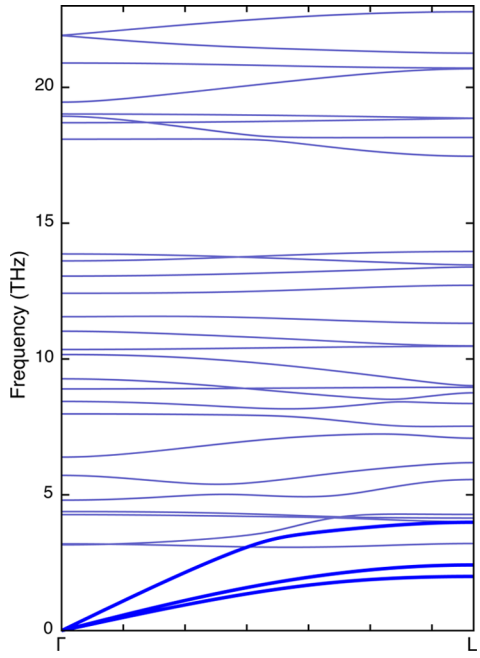


Figure 3. Phonon dispersion relationship along the first Brillouin zone of the β-Ga_2O_3 primitive cell from DFT calculations. The acoustic branches are highlighted.

β-Ga_2O_3 from Γ to L corresponding to the $(\bar{2}01)$ plane are used for the TBC calculations. The corresponding phonon group velocities can be found in the Supporting Information.

3. RESULTS AND DISCUSSION

3.1. TBC at β-Ga_2O_3/Metal Interfaces. Both theoretical and experimental methods are applied in our work to explore the TBC at metal/β-Ga_2O_3 interfaces. For the theoretical predictions, with the phonon dispersion relation of both β-Ga_2O_3 and metal, the TBC can be obtained from the Landauer formula. The calculations of TBC as a function of temperature from 0 to 500 K are shown in Figure 4. It is found that, for all the metal/β-Ga_2O_3 interfaces, the TBC increases with the temperature because more phonon modes are activated with increasing temperature from the Bose−Einstein distribution and the total heat flux increases. As the temperature increases, the rate of the increase in TBC becomes smaller and the TBC starts to level off at a higher temperature. The levelling-off of the TBC occurs as the temperature approaches the Debye temperature of the metal. It is the metal Debye temperature that mainly affects the TBC because the cutoff frequency and Debye temperature of metal are much smaller than those of β-Ga_2O_3 as shown in Table 1. The phonon transmission coefficient across the metal/β-Ga_2O_3 interfaces from DMM assumes that there is only elastic phonon transport at the interface and that the phonon frequency does not change after the transmission process. As a result, high-frequency phonon modes of β-Ga_2O_3 above the metal cutoff frequency cannot transmit or contribute to the TBC. Consequently, the TBC is limited by the cutoff frequency and Debye temperature of the metal. We think that the elastic phonon transport assumption

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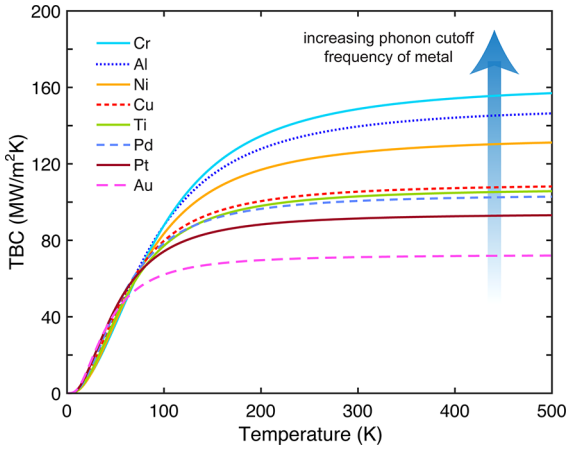


Figure 4. Temperature-dependent thermal boundary conductances (TBCs) at different β-Ga_2O_3/metal interfaces from 0 to 500 K predicted from the Landauer formula.

applied here is reasonable because, in a recent study of TBC at the Al/sapphire interface,[18] it is found that the inelastic scattering is negligible for the ultraclean metal/nonmetal interface.

The values and temperature dependence of TBC for different metals are similar from 0 to 60 K. The reason for the similarity is that the phonon properties of metals are relatively similar to those of β-Ga_2O_3. All the metals in our study except Ti have the FCC crystal structure with one atom in the FCC lattice primitive cell. The phonon dispersion curves of these FCC metals are very similar to one longitudinal acoustic (LA) curve and two degenerated transverse acoustic (TA) curves, while at the same time there are 30 phonon branches for β-Ga_2O_3, which has a complicated primitive cell with 10 atoms. Also, the cutoff frequencies of metals are all below 10 THz, while the cutoff frequency of β-Ga_2O_3 is 22.8 THz as shown in Table 1. As a result, the calculated TBCs are very similar for different metal/β-Ga_2O_3 interfaces at low temperature. However, the phonon cutoff frequencies and Debye temperatures of these metals are different, and the TBC versus temperature curves start to separate as temperature increases. Around 60 K, the TBC versus temperature curve of the Au/β-Ga_2O_3 interface begins to plateau while others keep increasing. Consequently, the TBC at the Au/β-Ga_2O_3 interface is the lowest among all eight studied metal/β-Ga_2O_3 interfaces. The cutoff frequency of Au is 4.6 THz, and its Debye temperature is 162 K, which are much lower than those of other metals. Because of the elastic transmission assumption, phonons with frequencies above 4.6 THz cannot transmit in this Landauer with DMM calculations, and the low cutoff frequency would result in a low plateau temperature and low TBC. The room-temperature Au/β-Ga_2O_3 TBC predicted from the Landauer formula is 71 MW/(m^2 K).

The TBC trends of other metals start to separate at about 70 K, and the room-temperature TBC values range from 90 to 150 MW/(m^2 K). For the ohmic contact metals, Cr/β-Ga_2O_3 shows the highest room-temperature TBC of 149 MW/(m^2 K), while the TBC at the most common Ti/β-Ga_2O_3 contact is 103 MW/(m^2 K). Among all the Schottky contact metals, the TBC at the Ni/β-Ga_2O_3 interface is the highest, which is 126 MW/(m^2 K) at room temperature, while the Au/β-Ga_2O_3 interface has the lowest TBC. Because the crystal structures of most metals in our study are FCC and elastic interface transport is assumed, the TBC value between β-Ga_2O_3 and metal is directly related to the cutoff frequency of the metal. The room-temperature TBC values together with the metal cutoff frequencies are shown in Table 1, and it is found that the sequence of metal cutoff frequencies perfectly matches that of the TBC.

For a better understanding of the relation between the phonon cutoff frequency of metal and the TBC at the β-Ga_2O_3/metal interface, we have calculated the spectral accumulation of TBC at 300 K as shown in Figure 5. The

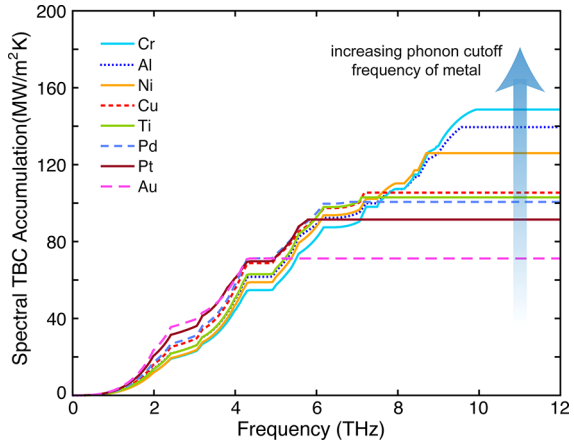


Figure 5. Spectral TBC accumulation of β-Ga_2O_3/metal interfaces at 300 K from the Landauer formula.

spectral accumulation shows that the TBC increases steadily with the frequency until the metal cutoff frequency, and the increasing rates are similar for different β-Ga_2O_3/metal interfaces. The spectral TBC accumulation demonstrates our hypothesis that the TBC between β-Ga_2O_3 and metal is dependent on the cutoff frequency of the metal. At the metal cutoff frequency, because of the elastic transport assumption, the TBC stops increasing with frequency. As a result, the TBC at the interface between β-Ga_2O_3 and metal with a lower cutoff frequency would stop increasing earlier and have a lower TBC. The TBC value is directly related to the phonon transmission coefficient across the interface, and the transmission as a function of frequency at different β-Ga_2O_3/metal interfaces can be found in the Supporting Information. From the spectral TBC accumulation, it is clear that, for the eight β-Ga_2O_3/metal interfaces that we study, the TBC at the β-Ga_2O_3/metal interface depends on the metal cutoff frequency.

3.2. Comparison of TBCs between Modeling and Experiments. The realistic metal/β-Ga_2O_3 contacts are very

Table 1. Phonon Cutoff Frequencies of β-Ga_2O_3 and Eight Metals Together with the Corresponding β-Ga_2O_3/Metal TBC at 300 K Calculated from the Landauer Approach with DMM

material	Ga_2O_3	Au	Pt	Pd	Ti	Cu	Ni	Al	Cr
cutoff frequency (THz)	22.8	4.4	5.8	6.8	7.1	7.2	8.7	9.6	9.9
TBC (MW/(m^2 K))	N/A	71.2	91.4	100.6	103.0	105.5	126.0	139.6	148.7

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Table 2. Room-Temperature TBC Results at Metal/β-Ga_2O_3 Interfaces from Modeling and Experiments[a]

Interface		Au/Ga_2O_3	Ti/Ga_2O_3	Ni/Ga_2O_3	Al/Ga_2O_3	Cr/Ga_2O_3
TBC from modeling (MW/m^2K)		71.2	103.0	126.0	139.6	148.7
TBC from experiments (MW/m^2K)	Our work	31.2 ± 2.8	17.4 ± 1.6	82.7 ± 7.4	81.7 ± 7.4	
	Ref. [7]				30 - 150	
	Ref. [10]	45	223	411		451

[a]Our modeling results with the Landauer method are shown in blue, and our experimental measurements with TDTR are shown in green.[7,10]

Table 3. Standard Molar Gibbs Free Energies of Formation $\Delta_f G^o$ Referring to 1 mol of Metal Oxide and Gibbs Free Energies of Oxidation $\Delta_f G^o(O_2)$ at 298.15 K Referring to 1 mol of O_2[a]

Metal oxide	Ga_2O_3	CuO	Cu_2O	NiO	Cr_2O_3
$\Delta_f G^o$ (kJ/mol)	**-998.3**	-129.7	-146	-212.65	-1058.1
$\Delta_f G^o(O_2)$ (kJ/mol O_2)	**-665.5**	-259.4	-292	-425.3	-705.4
Metal oxide	TiO_2	Ti_3O_5	Ti_2O_3	TiO	Al_2O_3
$\Delta_f G^o$ (kJ/mol)	-888.8	-2317.4	-1434.2	-495	-1582.3
$\Delta_f G^o(O_2)$ (kJ/mol O_2)	-888.8	-927	-956.1	-990	-1054.9

[a]Blue $\Delta_f G^o(O_2)$ means that the free energy of the metal oxide referring to 1 mol of O_2 is lower than that of β-Ga_2O_3, and there would be chemical reaction at the corresponding β-Ga_2O_3/metal interfaces.

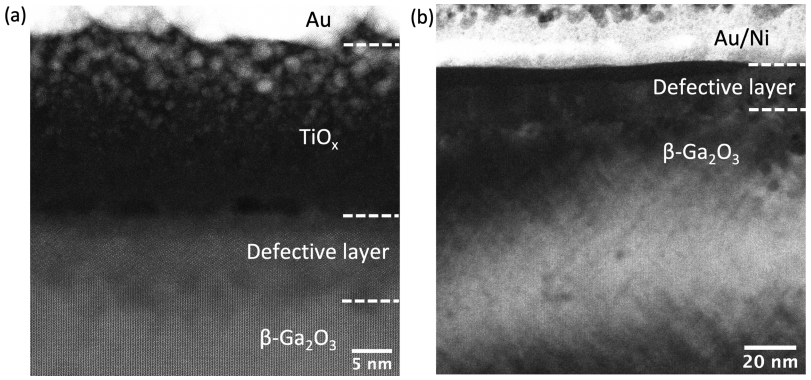


Figure 6. (a) STEM characterization of Au/Ti/β-Ga_2O_3 contact. (b) STEM characterization of Au/Ni/β-Ga_2O_3 contact. There is a chemical reaction at the Ti/β-Ga_2O_3 interface, and there are defective Ga_2O_3 layers for both interfaces.

complicated systems, and not all features of real interfaces can be captured in modeling. A comparison with experimental measurements is necessary to see if the cutoff frequency is still the main factor affecting TBC at real interfaces. A TDTR measurement of TBC has been applied to Al/β-Ga_2O_3, Au/β-Ga_2O_3, Ti/β-Ga_2O_3, and Ni/β-Ga_2O_3 interfaces with the $(\bar{2}01)$-oriented β-Ga_2O_3 substrate. For both Al/β-Ga_2O_3 and Au/β-Ga_2O_3 samples, thin-layer metals are E-beam evaporated β-Ga_2O_3 $(\bar{2}01)$ substrate. For the Ti/β-Ga_2O_3 and Ni/β-Ga_2O_3 samples, 20 nm layers of Ti and Ni are E-beam evaporated on top of the β-Ga_2O_3 $(\bar{2}01)$ substrate, respectively. On top of the Ti and Ni layers, a thin layer of Au is evaporated to imitate real device contact and act as the transducer for TDTR measurement. The details of TDTR measurements can be found in the Supporting Information Section S5. Although Au/Ni/β-Ga_2O_3 and Au/Ti/β-Ga_2O_3 contacts have several layers of metal, they are considered as Ni/β-Ga_2O_3 and Ti/β-Ga_2O_3 interfaces because the thermal resistance from the metal/metal interface and metal itself should be relatively small compared with metal/β-Ga_2O_3 thermal boundary resistance.

The measured TBCs at room temperature of the Au/β-Ga_2O_3, Ti/β-Ga_2O_3, Ni/β-Ga_2O_3, and Al/β-Ga_2O_3 interfaces (from low to high metal cutoff frequencies) are 31.2, 17.4, 82.7, and 81.7 $MW/(m^2\ K)$, respectively, as shown in Table 2. From the comparison between experiments and modeling, it is found that, for the relation between the TBC values of interfaces across β-Ga_2O_3 and different metals, there are some similar trends between modeling and experimental measurements. The TBC of Au/β-Ga_2O_3 interface is much lower than those of Al/β-Ga_2O_3 and Ni/β-Ga_2O_3 interfaces, and the TBCs of Al/β-Ga_2O_3 and Ni/β-Ga_2O_3 interfaces are similar, which are consistent with the modeling results. There is also a quantitative similarity between the relative TBC values from modeling and experiments. The Landauer based simulation predicts that the TBC of Au/β-Ga_2O_3 should be 51% of the TBC of Al/β-Ga_2O_3, and it was measured to be 38%, which is close to the predicted percentage from modeling. The smaller percentage of Au/β-Ga_2O_3 from experiments might come from the weak electron-phonon coupling and the bad adhesion of Au on top of Ga_2O_3 compared with Al and Ni.

Although there are some similarities between the modeling and experimental TBC results, there are some major differences. The experimentally measured TBC values from TDTR are generally lower than the modeling results, which indicates lower transmission than DMM predicted at real interfaces. Moreover, the measured TBC at the Ni/β-Ga_2O_3 interface is significantly lower than those of other interfaces despite the relatively high cutoff frequency of Ti, which is not captured from modeling. Since a clean interface between two

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crystalline materials is assumed in our modeling, we suspect that there might be some differences between the real interfacial structure and our modeled system.

3.3. Chemical Reactions and the Defective Ga_2O_3 Layer at Metal/β-Ga_2O_3 Interfaces. As pointed out in a previous study of Ga_2O_3 interfaces,[10] chemical reactions happen at the metal/β-Ga_2O_3 interfaces considering the relatively high Gibbs free energy of β-Ga_2O_3.[32,33] The standard molar Gibbs free energies of formation $\Delta_f G^o$ at 298.15 K for different materials related to β-Ga_2O_3/metal contacts are shown in Table 3. The Gibbs free energies of the oxides of Pt, Au, and Pd are not shown in the table because they have very high or even positive Gibbs free energies and will not react with β-Ga_2O_3. It should be noted that, for comparison of different reactions, all values of $\Delta_f G^o$ should refer to the reaction of the same quantity of oxygen as applied in the Ellingham diagram. Therefore, the Gibbs free energies of oxidation $\Delta_f G^o(O_2)$ referring to 1 mol of O_2 rather than 1 mol of the metal or metal oxide are also shown in Table 3. The lower the $\Delta_f G^o(O_2)$ value, the greater the stability of the corresponding metal oxide is, and the metal with the lower Gibbs free energy of oxidation will reduce the oxide with the higher Gibbs free energy of formation. As a result, Cr, Ti, and Al can reduce the oxide of β-Ga_2O_3 and there would be chemical reactions at the Cr-, Ti-, and Al-Ga_2O_3 interfaces, and there should not be chemical reactions at other studied metal/β-Ga_2O_3 interfaces (Au, Pt, Pd, Cu, and Ni).

Scanning transmission electron microscopy (STEM) has been used to further understand the interfacial structures and chemical reactions at the Ti/β-Ga_2O_3 and Ni/β-Ga_2O_3 interfaces as shown in Figure 6 (the growth direction of Ga_2O_3 is [010]). The evidence of chemical reaction is shown at the Ti/β-Ga_2O_3 interface (Figure 6a) but not at the Ni/β-Ga_2O_3 interface (Figure 6b), which is consistent with the Gibbs free energy analysis explained above. At the Ti/β-Ga_2O_3 interface, TiO_x is apparently formed in the whole evaporated Ti layer. The thermal conductivity of amorphous TiO_x is much lower than that of Ti, and the STEM result might explain the much lower TBC across the Ti/β-Ga_2O_3 interface from experimental measurements (Table 2). Moreover, there is a defective Ga_2O_3 layer at both Ni/β-Ga_2O_3 and Ti/β-Ga_2O_3 contacts although there is no chemical reaction at the Ni/β-Ga_2O_3 interface. The formation of the defective layer might be related to the growth kinetics.

To understand the difference between modeling and experiments quantitatively, the measured total thermal boundary resistance R_{total} (inverse of measured TBC) has been divided into two parts, the resistance of the 23 nm TiO_x layer R_{TiO_x} and the resistance at the Ti/β-Ga_2O_3 interface R_{Ti/Ga_2O_3}, and $R_{total} = R_{TiO_x} + R_{Ti/Ga_2O_3}$. The thermal conductivity of amorphous TiO_2, which is 1.33 W/(m K),[34] has been used to estimate R_{TiO_x}. TiO_2 is chosen due to the fact that the oxygen atomic fraction is about 2 times that of titanium from the atomic fraction of titanium and oxygen atoms. The resistance R_{Ti/Ga_2O_3} at the interface is 9.7×10^{-9} (m^2 K)/W from Landauer modeling. From the thermal conductivity estimation and thickness of the TiO_x layer, the resistance R_{TiO_x} is 17.3×10^{-9} (m^2 K)/W, which is nearly 2 times the R_{Ti/Ga_2O_3}, and the total resistance and the overall TBC are 2.7×10^{-8} (m^2 K)/W and 37 MW/(m^2 K), respectively. With the R_{TiO_x} of the TiO_x layer being considered, the inconsistency between modeling and experiments can be explained and the predicted TBC value is now much closer to the measured one. The remaining difference might be from the thermal resistance of the defective Ga_2O_3 layer and the Au/TiO_x interface.

3.4. Other Factors that Affect TBC at Real β-Ga_2O_3/Metal Interfaces. Electron−phonon coupling in metals might give additional thermal resistance for interfacial thermal transport at metal/β-Ga_2O_3 interfaces. The resistances for most metals with relatively high electron−phonon coupling factors would be negligible compared with the resistance from the phonon−phonon channel. However, for metals like Au and Cu with weak electron−phonon coupling strength values, the resistance would play a role in the overall interfacial resistance. The resistance from electron−phonon coupling can be estimated from the equation $R_{ep} = \left(\frac{k_e}{k_e + k_p}\right)^{3/2}\left(\frac{1}{G_{ep}k_p}\right)^{1/2}$.[35] For Au, the electron−phonon coupling factor G_{ep} is 2.6×10^{16} W/(m^3 K), and k_e and k_p are 311.4 and 2.6 W/(m K), respectively.[36] As a result, R_{ep} is 3.8×10^{-9} (m^2 K)/W. Together with the resistance from Landauer calculations of 14×10^{-9} (m^2 K)/W (the multiplicative inverse of the 71.2 MW/(m^2 K) conductance), the overall resistance would be 17.8×10^{-9} (m^2 K)/W, which means that the corresponding TBC considering the electron−phonon coupling would be 56.2 MW/(m^2 K), and it is closer to our TDTR measurement. For Cu, G_{ep} is 5.55×10^{16} W/(m^3 K), and k_e and k_p are 384 and 17 W/(m K), respectively.[37,38] As a result, R_{ep} is 9.6×10^{-10} (m^2 K)/W. Together with the resistance from Landauer calculations of 9.48×10^{-9} (m^2 K)/W (105.5 MW/(m^2 K) conductance), the overall resistance would be 10.44×10^{-9} (m^2 K)/W, which means the corresponding TBC would be 95.8 MW/(m^2 K).

Our modeling and experimental TBC results are also compared with previous published results. In a recent study with TDTR of the thin-film β-Ga_2O_3 thermal conductivity by Zhang *et al.*,[7] the measured TBC at the Al/β-Ga_2O_3 interface ranges from 30 to 150 MW/(m^2 K). The TBC range from their measurements is consistent with our measurement of 81.7 MW/(m^2 K). In another study of the metal/β-Ga_2O_3 interface by Aller *et al.*,[10] the measured TBC at the Au/β-Ga_2O_3 interface is about 45 MW/(m^2 K), which is higher than but similar to our measured value of 31.2 MW/(m^2 K).

Aller *et al.*[10] also measured the TBC values at Au/Ni/β-Ga_2O_3, Au/Ti/β-Ga_2O_3, and Au/Cr/β-Ga_2O_3 interfaces, and the TBC values are about 411, 223, and 451 MW/(m^2 K), respectively, as shown in Table 2. It should be noted that they studied the thickness-dependent TBC, and the values shown here are the plateau TBC at large thickness. As mentioned above, although Au/Ni/β-Ga_2O_3, Au/Ti/β-Ga_2O_3, and Au/Cr/β-Ga_2O_3 contacts have several layers of metal, they are considered as Ni/β-Ga_2O_3, Ti/β-Ga_2O_3, and Cr/β-Ga_2O_3 interfaces. When compared with our modeling results, the sequence of their measurements is $TBC_{Cr} > TBC_{Ni} > TBC_{Ti} > TBC_{Au}$, which matches our modeling prediction exactly. Here, TBC_{metal} means the TBC at a certain metal/β-Ga_2O_3 interface. When compared with our experimental results, there are several differences. One big difference is that the TBC at the Ti/β-Ga_2O_3 interface in ref[10] is not significantly lower than that of Ni/β-Ga_2O_3 interfaces. Another big difference is that the TBC values in ref[10] are generally higher than our measurements. In fact, the TBC values in ref[10] are higher

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than our modeling results, and the reason would be further discussed in Section 3.5.

There are several factors that might lead to the different measured TBC values at the Ti/β-Ga_2O_3 interface. Ref[10] applied thermal evaporation at room temperature rather than our E-beam evaporation method for metal deposition, and the vacuum level is higher than what we apply in our experiment. The ultrahigh vacuum level together with the difference in growth kinetics might lead to a thinner oxidation layer and as a result higher TBC values for metal/β-Ga_2O_3 contacts with chemical reactions. The reason for the generally higher TBC values in ref[10] might come from different β-Ga_2O_3 orientations. The β-Ga_2O_3 orientation is $(\bar{2}01)$ for our measurements, while the β-Ga_2O_3 orientation is (100) in ref[7] and (010) in ref[10] There is strong anisotropy in β-Ga_2O_3 because of its monoclinic structure, and it would be reasonable if there is an anisotropy effect in TBC as well.

From the comparison between different experiments and modeling of TBC at metal/β-Ga_2O_3 interfaces, it is found that the conditions at real interfaces are extremely complicated and there are many factors that affect the TBC, such as different metals, chemical reactions, growth methods, conditions and kinetics, electron−phonon coupling, and Ga_2O_3 orientations. Because of these complexities at real interfaces, it is reasonable that there would be discrepancy between the TBC values in different experiments and also theoretical modeling and experimental measurements. In fact, although there are differences between the TBC values from modeling and experiments, the experimental and modeling results generally follow a similar trend. The metal with a larger cutoff frequency usually has a larger TBC at β-Ga_2O_3/metal interfaces unless there would be intensive oxidation and chemical reaction at the interface. The chemical reactions, although cannot be avoided, might be controlled by growth methods and conditions like the vacuum level.

3.5. Comparison of Modeling Assumptions with Realistic Scenarios. From the comparison of TBC values from our modeling and experiments, it is found that there are differences between modeling results and experimental measurements. The difference of TBC between different experiments might come from growth methods, β-Ga_2O_3 orientations, electron−phonon coupling, and interface chemical reactions as discussed in prior sections. However, the difference between modeling and experiments might come from some modeling assumptions, which cannot fully capture the real interface.

To fully understand the difference in TBC values between modeling and experiments, we take a closer look at all the assumptions of the theoretical Landauer formula and compare them with realistic interfaces. It is found that there are several assumptions of the Landauer method, which might explain the difference. The main assumptions applied in our Landauer approach with transmission from DMM are elastic phonon transmission, local equilibrium near the interface, and isotropic material properties. These assumptions might lead to underestimation of the TBC from the Landauer formula with DMM.

At real β-Ga_2O_3/metal interfaces, inelastic scattering might contribute to the interfacial thermal transport, and as a result, our modeling with only an elastic transport channel might underestimate the TBC at the interface. The intensity of inelastic scattering at the interface is related to the growth method, interface defects, temperature, and many other factors. The inelastic scattering seems to be negligible at the ultraclean and harmonic metal/semiconductor interface.[18] However, it is hard to quantify the inelastic contribution at real interfaces especially the β-Ga_2O_3/metal interfaces, which involves complex chemical reactions. The local equilibrium assumption will also underestimate the TBC $(q/A\Delta T)$ at β-Ga_2O_3/metal interfaces[39−41] because the temperature drop across the interface ΔT is overestimated without considering the contribution of transmitted phonons to local temperature near the interface. Therefore, both inelastic scattering and local nonequilibrium effects at interfaces would lead to underestimation of TBC from Landauer modeling, which does not consider these effects.

Besides the elastic transmission and local equilibrium assumption, the shrunken β-Ga_2O_3 first Brillouin zone from the isotropic assumption would also lead to underestimation from modeling. β-Ga_2O_3 is an anisotropic monoclinic material, and the first Brillouin zone of the β-Ga_2O_3 primitive cell is also anisotropic as shown in Figure 7. Since the phonon dispersion

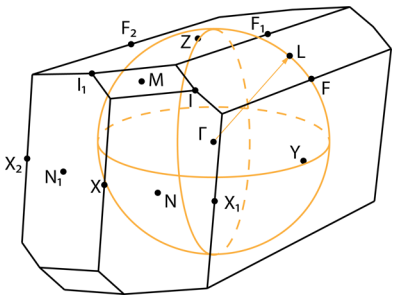


Figure 7. First Brillouin zone of β-Ga_2O_3 from its primitive cell and the integration volume applied in the Landauer formula with the isotropic assumption (orange ball).

from Γ to L perpendicular to the $(\bar{2}01)$ plane is applied in our calculation, the integration of the TBC over all frequencies with the isotropic assumption as shown in eqs 1 and 2 is in fact over the orange ball with a radius of $\Gamma\bar{L}$ in the *k*-space. The volume of the orange ball is only 53.67% of that of the β-Ga_2O_3 first Brillouin zone, and the total heat flux across the β-Ga_2O_3/metal interface would be underestimated from the integration over the isotropic ball rather than the real first Brillouin zone of β-Ga_2O_3 as contribution from phonon modes in the remaining volume is neglected. The TBC contribution as a function of the wavevector can be found in the Supporting Information.

There are two possible methods for expanding the integration to include the volume of the whole first Brillouin zone. The first method is to expand both the wavevector and the phonon frequency, to keep the phonon group velocity staying the same. The wavevector is extended from $\Gamma \rightarrow r$ to $_{BZ}$, where r_{BZ} is the effective radius of the β-Ga_2O_3 first Brillouin zone, that $V_{BZ} = \frac{4}{3}\pi {r_{BZ}}^3$ and V_{BZ} is the volume of the β-Ga_2O_3 first Brillouin zone. The second method is to expand just the wavevector but not the phonon frequency, to keep the phonon cutoff frequency of β-Ga_2O_3 staying the same. The expanded phonon dispersion relation of these two methods can be found in the Supporting Information. For our β-Ga_2O_3/metal interfaces, the frequencies of phonons in metal are generally low, so the first method should work better considering that the increased β-Ga_2O_3 cutoff frequency would not contribute to the TBC as the TBC is limited by the low metal cutoff frequency.

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We have applied the first expansion method to cover the whole first Brillouin zone volume, and the interface temperature correction[41] for the local nonequilibrium effect near the interface is also included for better accuracy of the Landauer formula. The TBC modeling results with Brillouin zone expansion and nonequilibrium correction are shown in Figure 8. It can be seen that, with both corrections, the TBC results at

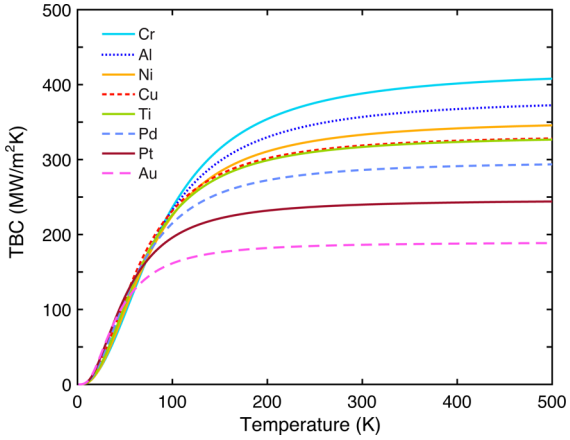


Figure 8. Temperature-dependent thermal boundary conductances (TBCs) at different β-Ga_2O_3/metal interfaces from 0 to 500 K predicted from the Landauer formula with isotropic Brillouin zone expansion to r_{BZ} and the nonequilibrium interface temperature correction.

all interfaces increase, which is reasonable because both the shrunken Brillouin zone and neglecting interface nonequilibrium would lead to underestimation of the TBC. What should be noted here is that the relation of TBC increasing with metal cutoff frequency still holds and the general TBC versus T trend is similar.

4. CONCLUSIONS

To summarize, the thermal boundary conductances at eight β-Ga_2O_3/metal interfaces were studied with the Landauer formula and TBCs at four of them were measured with TDTR. To understand the thermal transport at β-Ga_2O_3 interfaces, the modeling and experimental results were compared with each other and were analyzed from metal properties, interface conditions, and modeling assumptions. From our study, it has emerged from the modeling that the TBC value sequence follows the metal cutoff frequency sequence, and experimental measurements are generally consistent with the modeling observation except for the Ti/Ga_2O_3 interface with ultralow TBC. The low TBC at the Ti/Ga_2O_3 interface can be explained from the strong chemical reactions and the thick TiO_x layer. The generally similar trends of TBC and metal cutoff frequencies came from the similar properties of metals and the large difference between the phonon dispersion relations of metals and β-Ga_2O_3. It is found that, among all ohmic contact metals, the Cr/β-Ga_2O_3 interface had the highest TBC value, and for Schottky contact metals, the Ni/β-Ga_2O_3 interface had the highest TBC value. It was revealed that, although there are many factors that affect the β-Ga_2O_3/metal TBC at real interfaces, the metal cutoff frequency plays an important role. Our results could be applied as guidance and input for the thermal management design of β-Ga_2O_3 devices.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.1c05191.

Landauer approach modeling process, phonon group velocity of β-Ga_2O_3 along Γ→L of the first Brillouin zone, phonon transmission coefficients across metal/β-Ga_2O_3 interfaces, TBC densities as a function of wavenumber k, expanded phonon dispersion of β-Ga_2O_3, and time-domain thermoreflectance measurement procedure (PDF)

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Notes

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